TRIGONOMETRIC RATIOS

PLANE TRIGONOMETRY

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Basic Concepts

Angle, Degrees, and Radians

Right Triangles

Hypotenuse, Opposite Side, and Adjacent Side

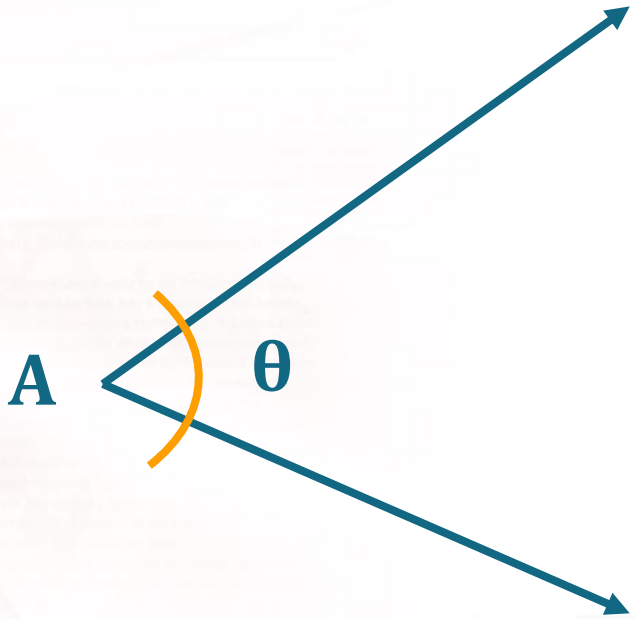
Trigonometric Ratios

sine, cosine, and tangent

BASIC CONCEPTS

ANGLE

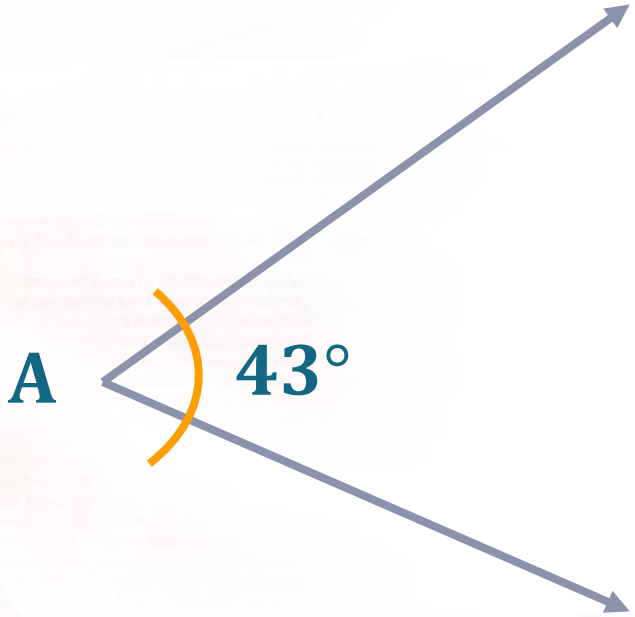
Formed by **two rays** or line segments that share a common endpoint, called the **vertex**



DEGREES

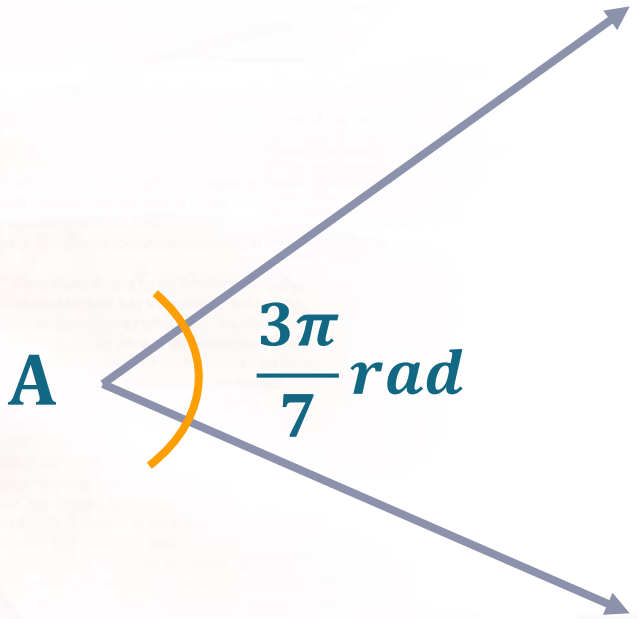
Unit of measurement for angles

Denoted by the symbol " ° "



Alternative unit of measurement for angles

Denoted by the symbol " **rad** "

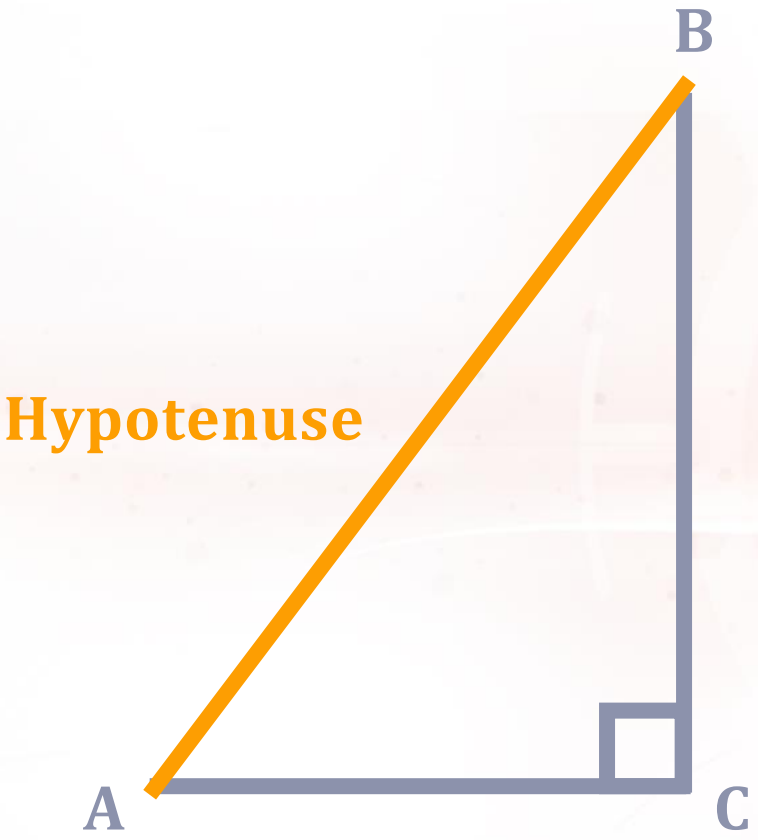


Convert the angle **60°** to **radians**.

Convert the angle $\frac{\pi}{4}$ **radians** to **degrees**.

INTRODUCTION TO RIGHT TRIANGLES

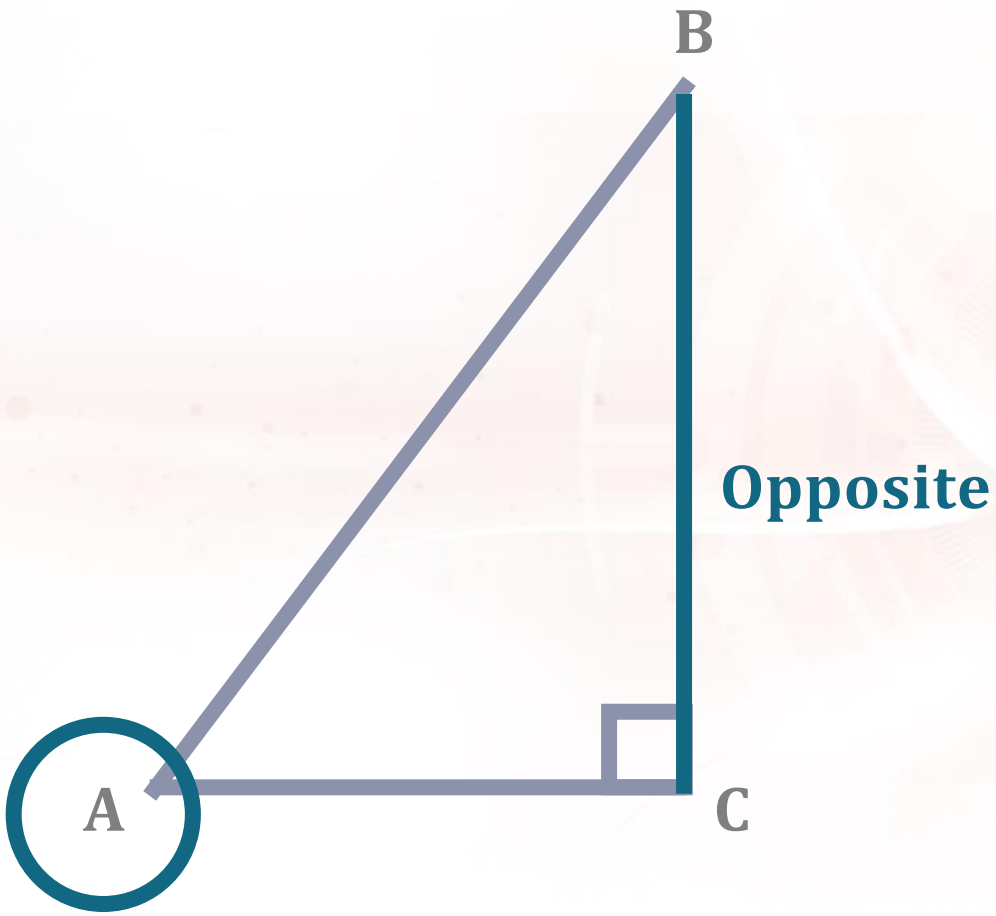
HYPOTENUSE



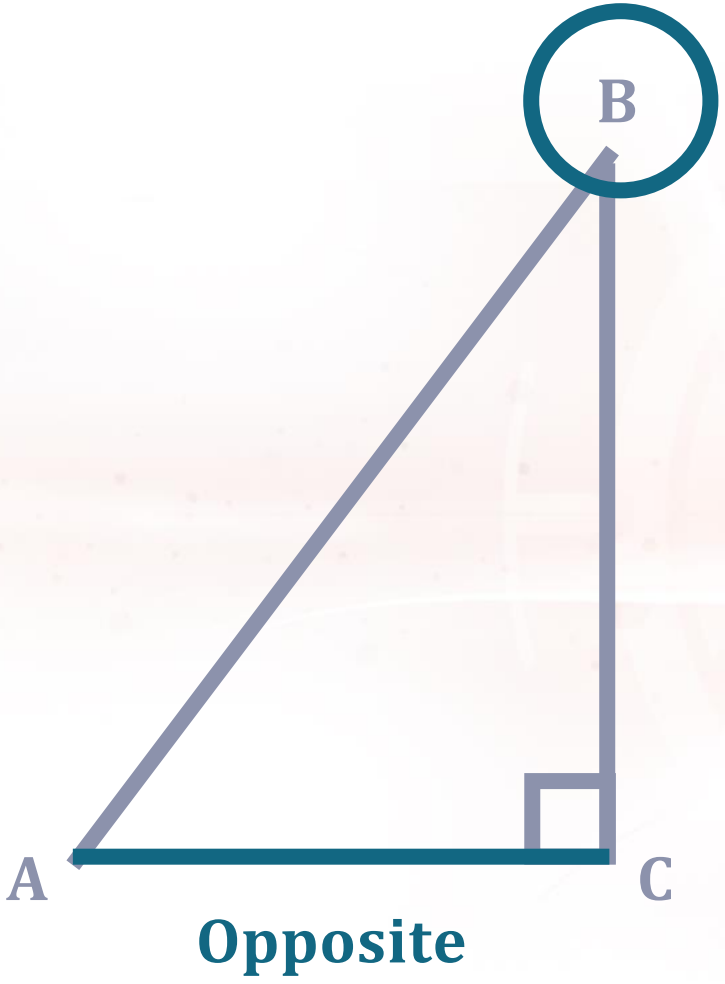
The **longest** side in a right triangle

The side opposite the right angle

OPPOSITE SIDE

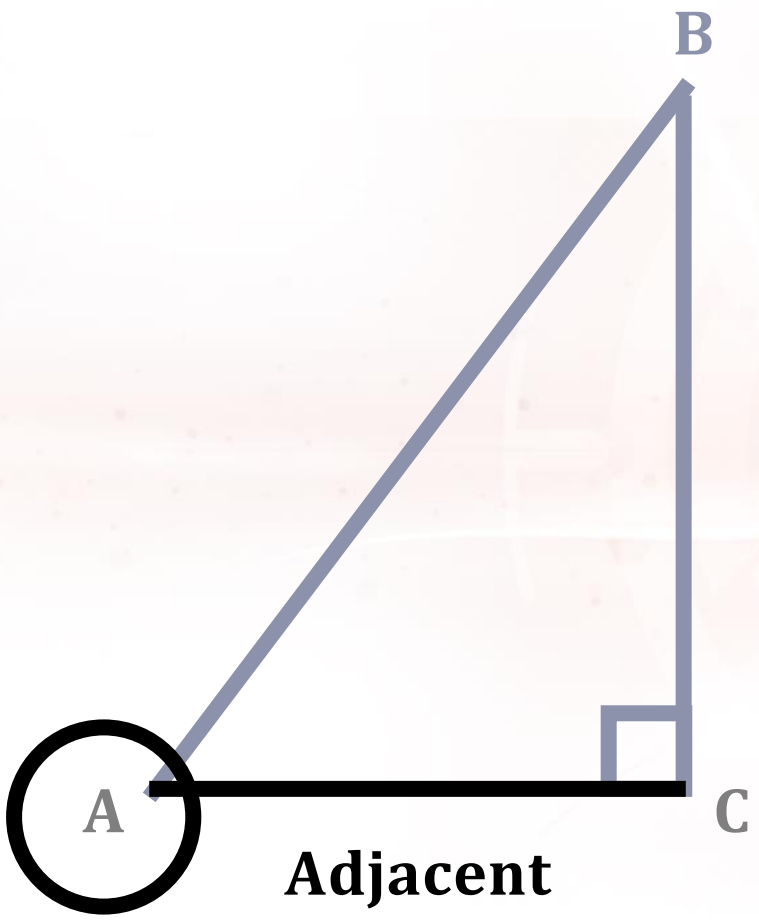


Across from a given angle



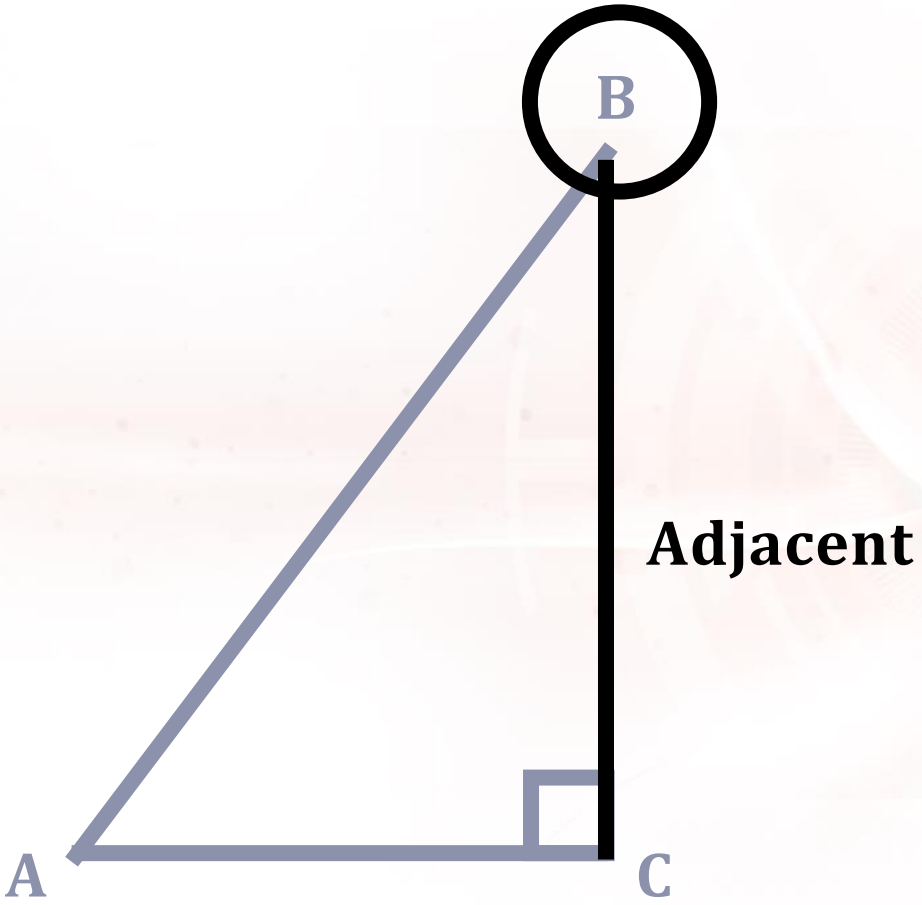
Across from a given angle

ADJACENT SIDE



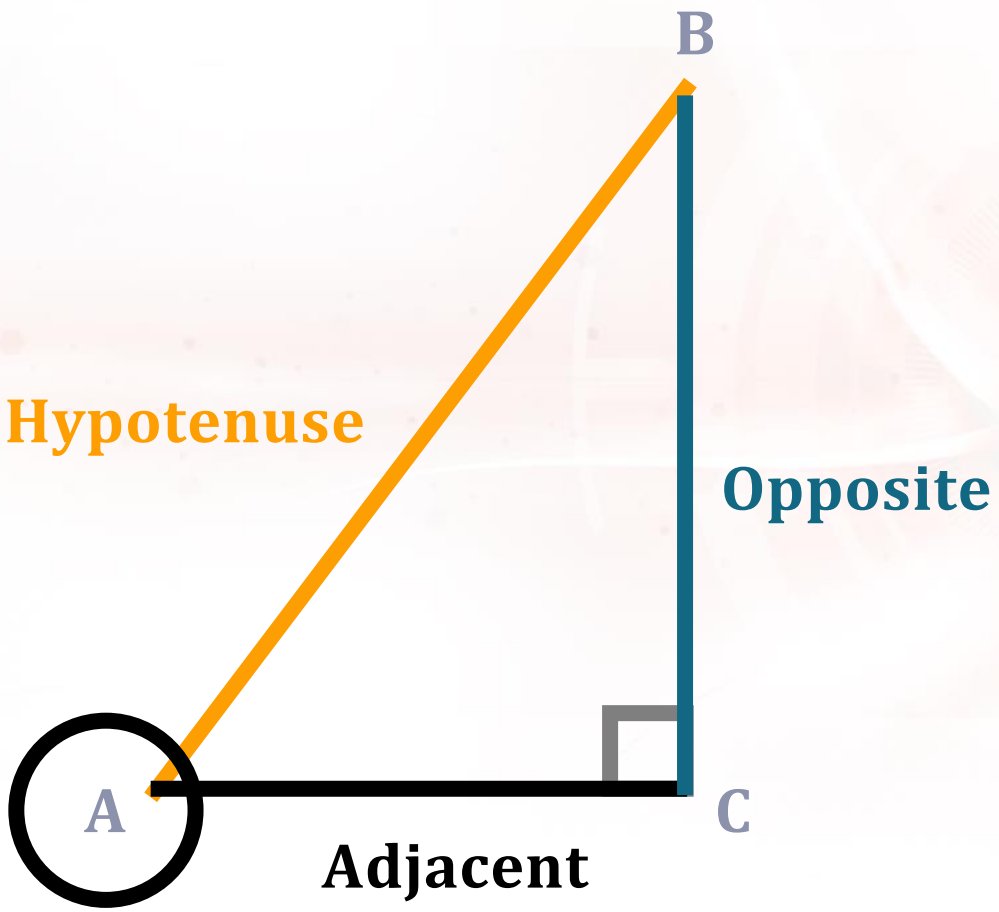
Next to a given angle

with respect to **ANGLE B**

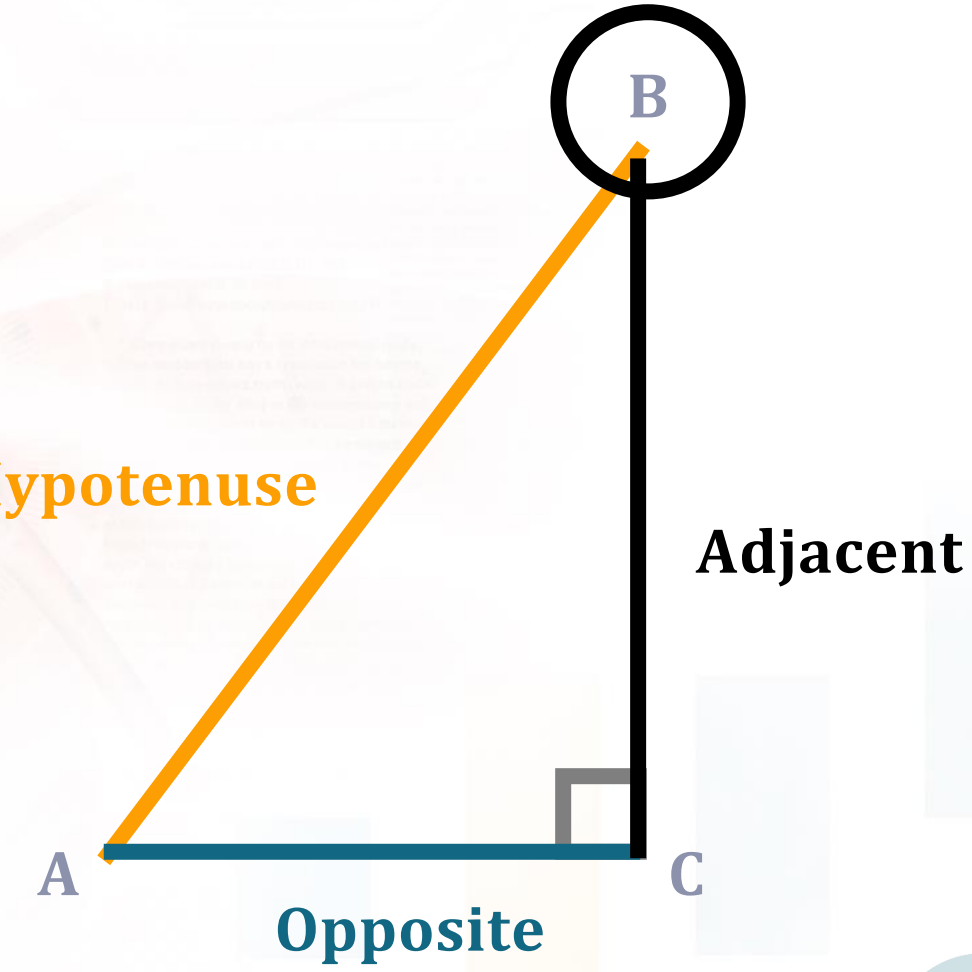


Next to a given angle

from the perspective of $< A$,



from the perspective of $< B$,



Relative to **angle G**, which side is the **adjacent** side △ EMG?

Relative to **angle N**, which side is the **opposite** side △ END?

TRIGONOMETRIC RATIOS

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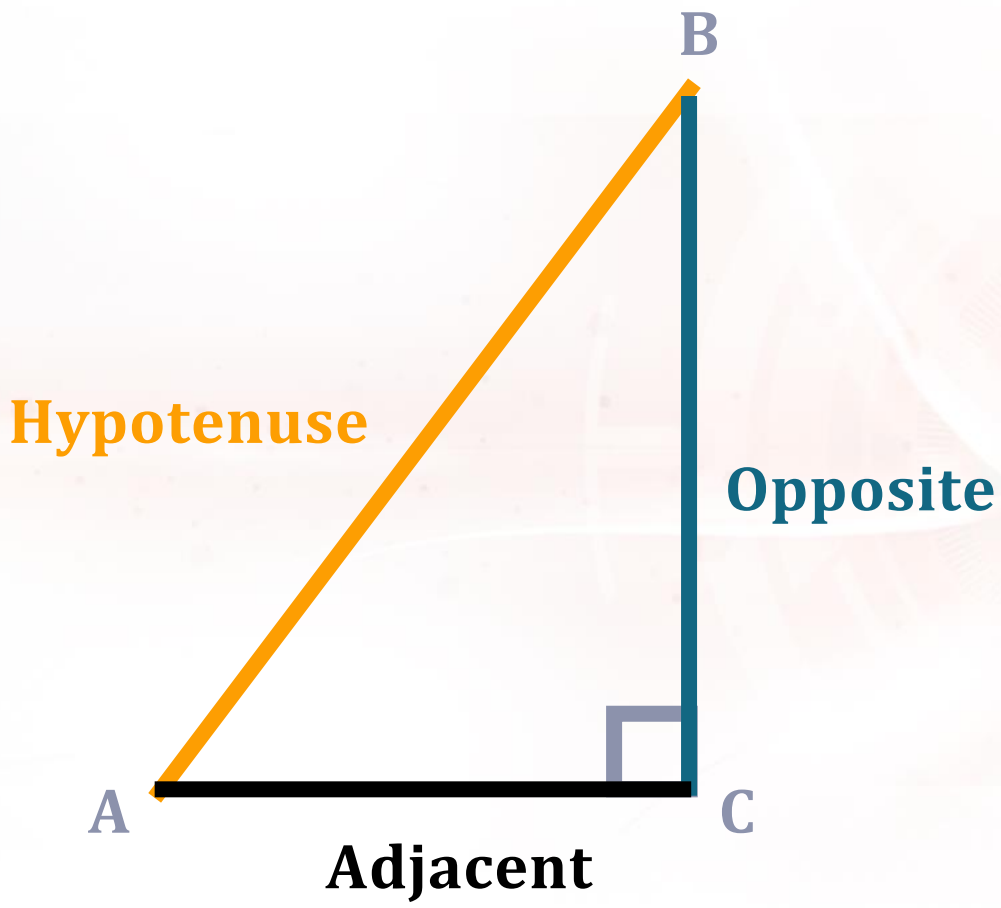
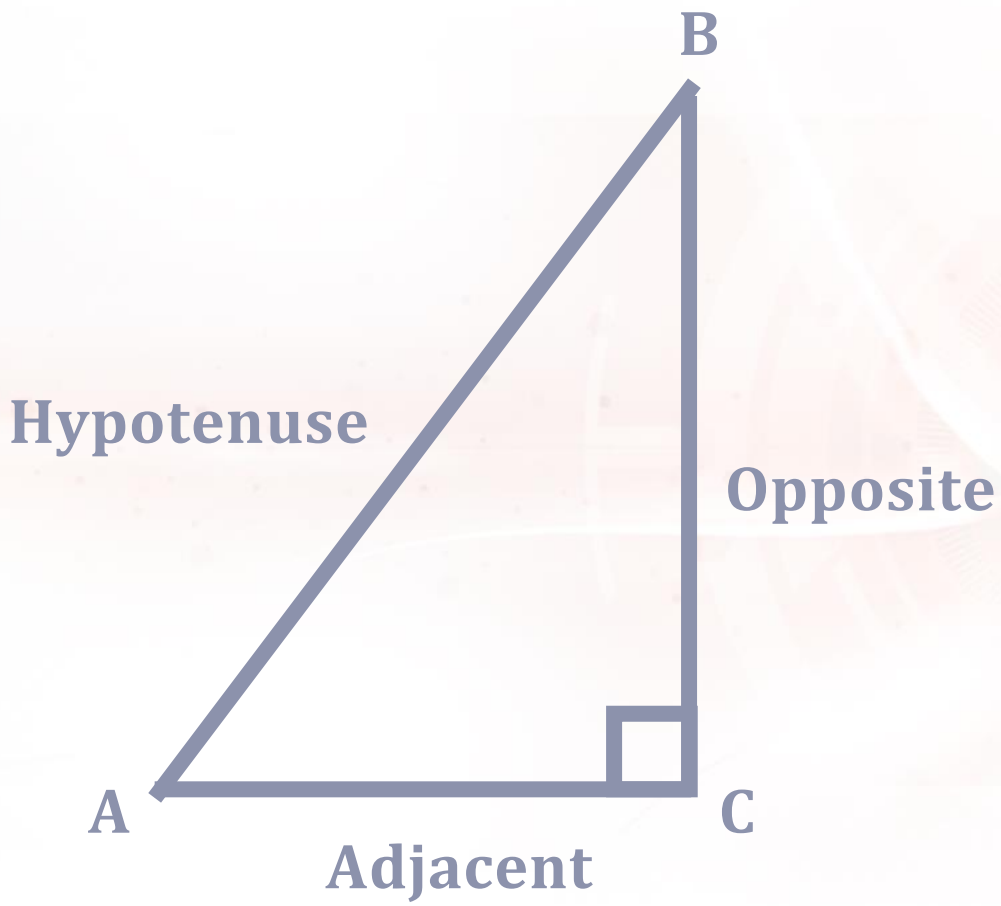


$$sin(A) = \frac{\textbf{Opposite}}{\textbf{Hypotenuse}} \qquad soh$$

$$cos(A) = \frac{\textbf{Adjacent}}{\textbf{Hypotenuse}} \qquad cah$$

$$tan(A) = \frac{\textbf{Opposite}}{\textbf{Adjacent}} \qquad toa$$

RECIPROCAL TRIGONOMETRIC RATIOS

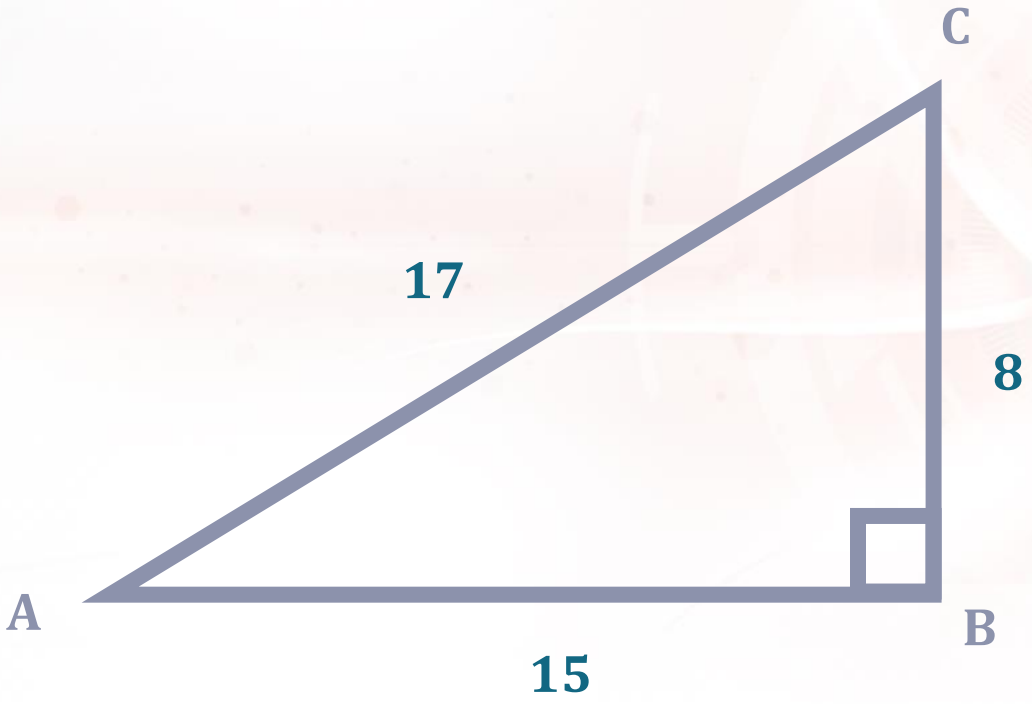


$$csc(A) = \frac{\text{Hypotenuse}}{\text{Opposite}} \qquad \frac{1}{sin(A)}$$

$$sec(A) = \frac{\text{Hypotenuse}}{\text{Adjacent}} \qquad \frac{1}{cos(A)}$$

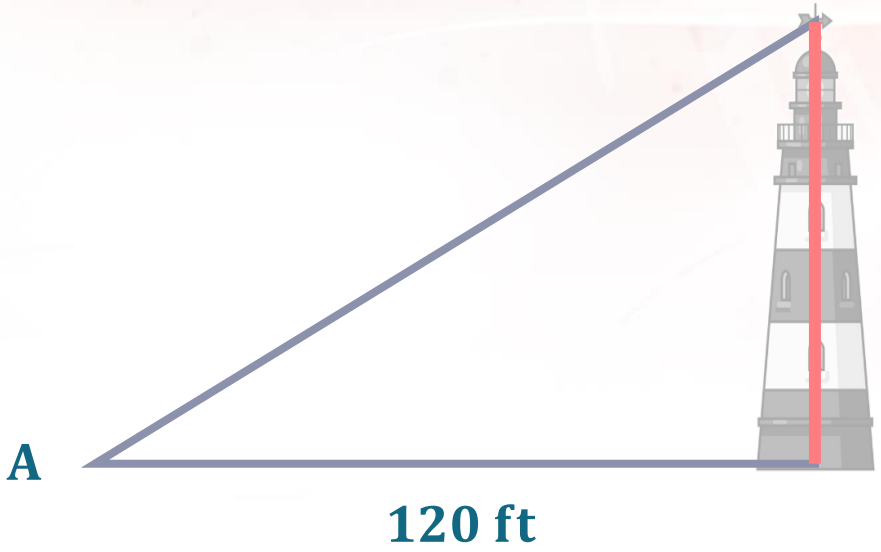
$$cot(A) = \frac{\text{Adjacent}}{\text{Opposite}} \qquad \frac{1}{tan(A)}$$

In the given **triangle**, find **sin A** , **cos C** , **csc A**, and **tan C**



A surveyor wishing to find the height of a lighthouse measures the angle A at a point **120** ft. from its base. His findings are represented in the given figure, where

$\boldsymbol{tan\,A} = \frac{2}{3}$. What is the **height** of the lighthouse?



SEATWORK